BIS108 BUSINESS REQUIREMENTS ANALYSIS | ASSESSMENT 4

Hartwell Primary School Sustainable Travel System

Boroondara Schools Project Prototype, CASE Tool Analysis & Usability Verification

Sofia Mironenko | Kim Le

Prototyping Approach Evaluation Matrix

Evaluation Criterion	Evolutionary (Selected)	Throwaway (Rejected)	Horizontal (Rejected)
Adaptiveness to Change	5/5: Modular updates	2/5: Rebuilt from zero	3/5: Surface labels only
End-to-End User Goal	5/5: Full task journeys	3/5: Isolated fragments	1/5: Superficial shells
Client Sign-Off Utility	5/5: Realistic controls	2/5: Low-fidelity sketch	2/5: Hides workflow logic
Iterative Alignment	5/5: Continuous baseline	3/5: Discovery spikes only	2/5: Premature breadth
Total Weighted Score	**20 / 20 (Optimal)**	**10 / 20 (Disqualified)**	**8 / 20 (Disqualified)**

Decisive Criterion: Adaptiveness to Change & Zero-Install Classroom Operability (Global colour swap across 9 screens in <10s).

Selection Justification & Risk Mitigation

Why Evolutionary Won

• Iterative Baseline: Refines high-priority safety workflows continuously.

• Throwaway Disqualified: Rebuilding wastes the $40k–$50k budget.

• Horizontal Disqualified: Shallow menus fail 500m hub check-ins.

• Goal Validation: Verifies end-to-end non-smartphone child tracking.

Identified Risk & Mitigation

• Core Risk: Scope creep from polished UI appearance.

• False Readiness: Mistaking front-end simulation for live backend.

• SRS Anchoring: Enforce strict SRS Use Case boundaries (UC-01–09).

• Change Control: Restrict edits to formal requests (CR-01).

CASE Tool Investigation & Decisive Criterion

CASE Tool Package	1. Cost	2. Learning Curve	3. Classroom Run	4. Adaptiveness	Total
Figma + Browser HTML (Selected)	**5/5 Free Edu**	**5/5 Intuitive UI**	**5/5 Zero-Install Browser**	**5/5 Global Tokens (<10s)**	**20 / 20**
Balsamiq Wireframes (Rejected)	3/5 30d Trial	5/5 Simple Sketch	3/5 Fails Colour Rubric	2/5 Manual Screen Edits	**13 / 20**
Axure RP 10 (Rejected)	2/5 $25–$49/m	2/5 Steep Logic	3/5 Heavy Local Install	4/5 Slow Team Sync	**11 / 20**

Kim

Colour Schemes Options

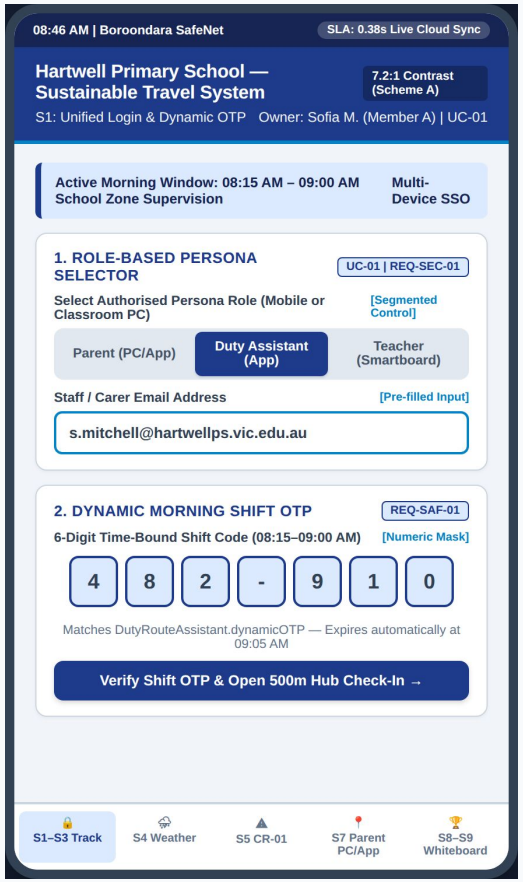
Attribute	Colour Scheme A: 'Municipal Corporate Navy' (Rejected)	Colour Scheme B: 'Boroondara Eco-Safe Green' (Client Selected)
Primary Brand Colour	Deep Navy (#1E3A8A)	Forest Emerald (#1B5E20)
Secondary / Action Accent	Sky Blue (#0284C7)	High-Visibility Teal (#00796B)
Surface & Background	Cool Grey (#F1F5F9) / White (#FFFFFF)	Warm Off-White (#F8FAFC) / Pure White Cards (#FFFFFF)
Semantic Status / Hazard Alerts	Standard Red (#EF4444)	High-Contrast Amber (#D97706) & Crimson (#B91C1C)
Typography & WCAG Contrast	Dark Slate (#334155) Contrast Ratio 7.2:1	Charcoal Ink (#111827) Contrast Ratio 11.4:1 (WCAG 2.2 AAA)

Kim

Client Colour Schemes & Change Request CR-01

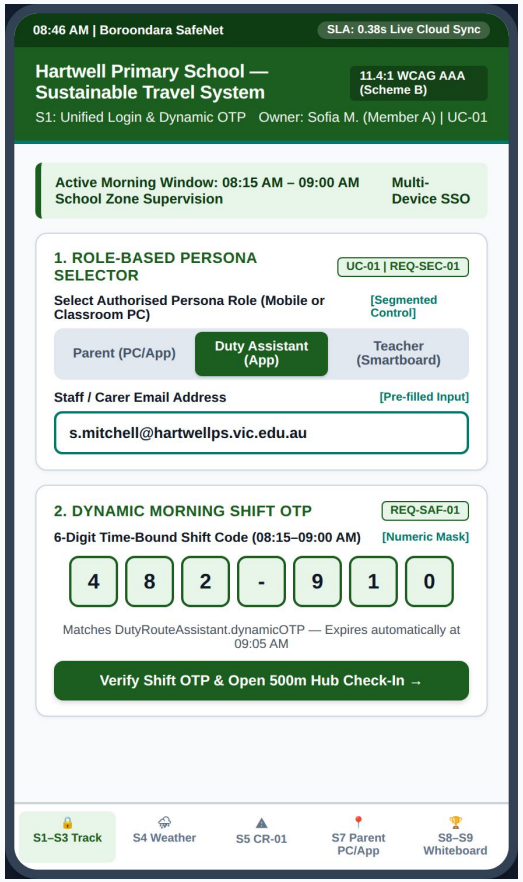
Scheme A: Municipal Navy

Corporate feel lacked environmental identity.



Scheme B: Eco-Safe Green

Outdoor daylight contrast



Colour Schemes psychology - Eco-Safe Green

Forest Emerald (#1B5E20)

- nature, growth, health and harmony
- calm, reassuring feeling for parents and students
- reinforces environmentally friendly

High-Visibility Teal (#00796B)

- trust and stability
- nature and wellbeing

Warm Off-White (#F8FAFC) &Pure White Cards (#FFFFFF)

- clean
- spacious appearance
- focus on important information

Crimson (#B91C1C)

- greater urgency and seriousness
- critical safety alerts

High-Contrast Amber (#D97706)

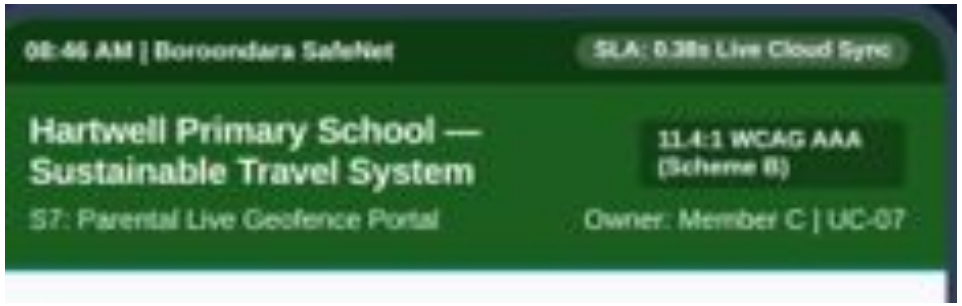
- caution and draws attention
- without necessarily indicating immediate danger

Charcoal Ink (#111827)

- strong contrast against white backgrounds
- clarity and professionalism
- making text easier to read

Prototype — iteration

First Iteration



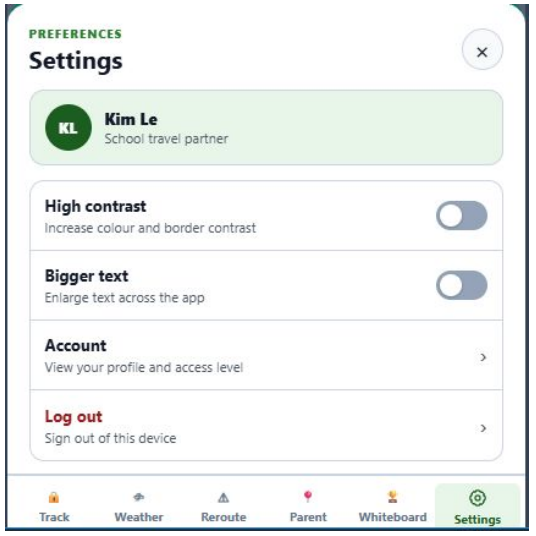


Prototype — iteration

Second Iteration

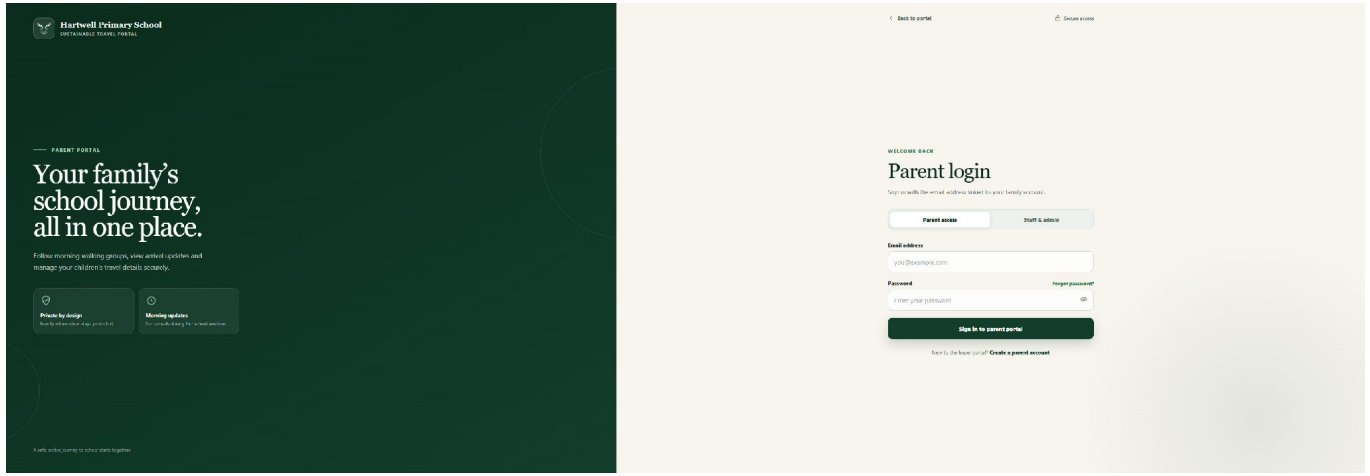


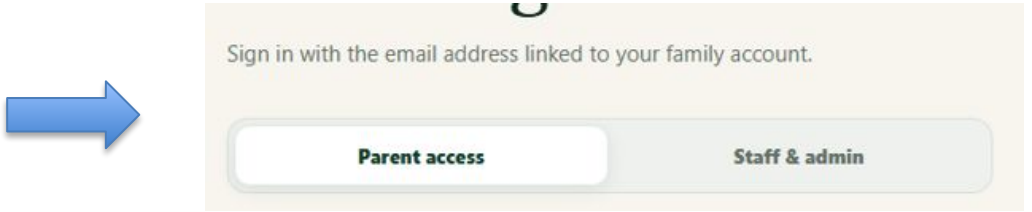


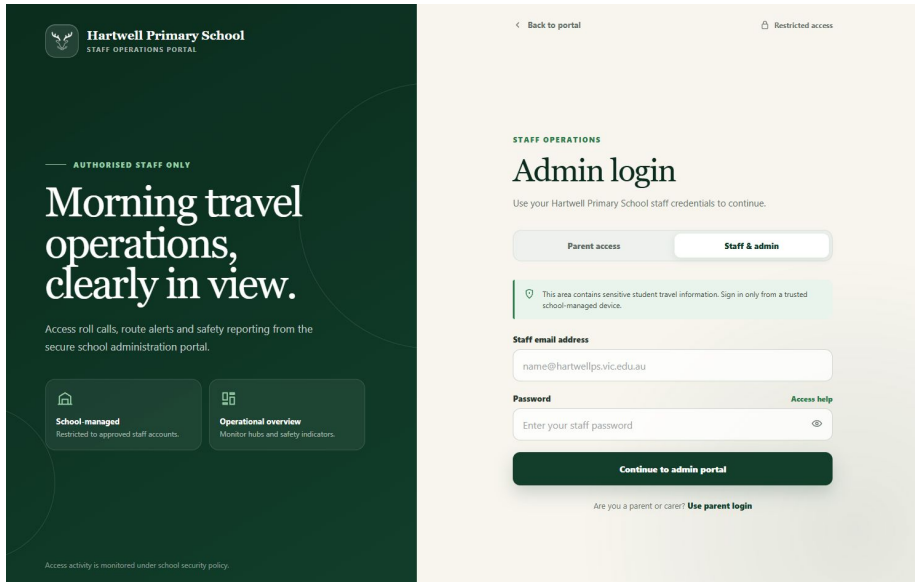


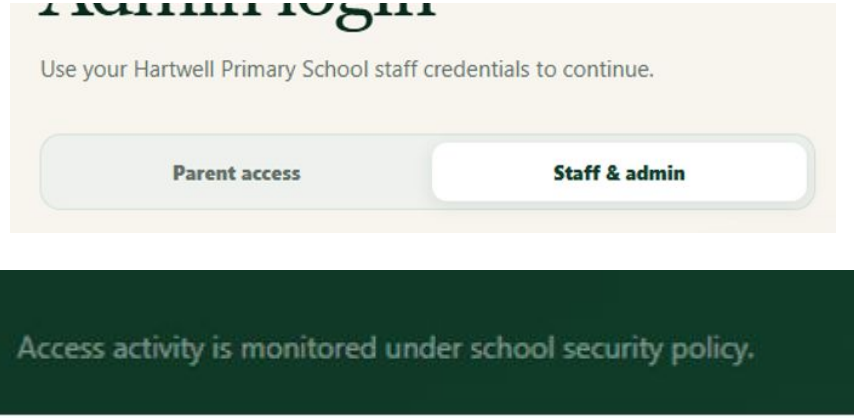
Prototype — iteration

Third Iteration









9-Screen SRS Traceability & Group Attribution

Subsystem Owner	Attributed Screens	SRS Use Cases & Req IDs	Matched UI Controls Implemented
Sofia Mironenko	S1 Login, S2 500m Check-In, S3 9AM SOS & Summary S7 Parent Map, S8 Privacy & Leaderboard, S9 Council Export	UC-01 to UC-03 | REQ-SEC-01, REQ-TRK-01/02, REQ-SAF-01/02 UC-07 to UC-09 | REQ-PAR-01, REQ-ETH-01, REQ-ACC-01	Segmented Role, 6-Digit OTP, Calendar Date Picker, 2-Step SOS Modal Sibling Dropdown, Time-Bounded GPS Window, Locked PII Toggle, 125% Slider
Kim Le	S4 BOM Feed, S5 CR-01 Override, S6 Hub Roll Call & KPIs	UC-04 to UC-06 | REQ-ENV-01/02, REQ-TRK-03, REQ-HAZ-01/02	Corridor Dropdown, Constrained CR-01 Detour Dropdown, 1:1 Check, KPI Gauges

Nielsen Usability Heuristics

3 Named Heuristics (Nielsen, 1994)

1. Visibility of System Status (#1): Live 14/15 check-in counter (S2), 0.38s SLA pill, 08:46 AM arrival banner (S7).

2. Error Prevention (#5): Constrained 500m hub dropdowns (S2, S5), two-step SOS modal (S3), locked PII toggle (S9).

3. Consistency & Standards (#4): Persistent 9-screen menu, uniform field labels, 11.4:1 WCAG AAA contrast.

Peer Usability Testing Log & References

Reviewer	Target Screen Tested	Usability / Ethical Defect Found	Fix Implemented in Final Build
Sofia M	S5 Override	Free-text detour input risked typos in rain Single-tap SOS button risked false parent panic	Constrained 500m Hub dropdown + 1:1 check Two-step Error Prevention confirmation modal
Kim Le on computer	S8 & S9 Export	Unchecking PII anonymization & 24/7 GPS risked APP 6 breach	Locked-ON PII toggle + 08:00–09:00 AM window

LIVE DEMONSTRATION

Prototype Walkthrough

https://www.figma.com/proto/SEaREimZuszWhDAjzXqVbF/draft?node-id=0-1&t=punGeqvbqjvprR8Z-1